We originally introduced the concept of the definite integral to calculate the area of a region bounded by a function $f(x) \geq 0$, the x-axis, and the vertical lines $x = a$ and $x = b$. We now expand that idea to calculate the area of more complex regions.

1 Area of a Region Between Two Curves

Let $f(x)$ and $g(x)$ be continuous over an interval $[a,b]$ such that $f(x) \geq g(x)$ on $[a,b]$. We want to find the area of the region S between the graphs of the functions, as shown in the figure below. We

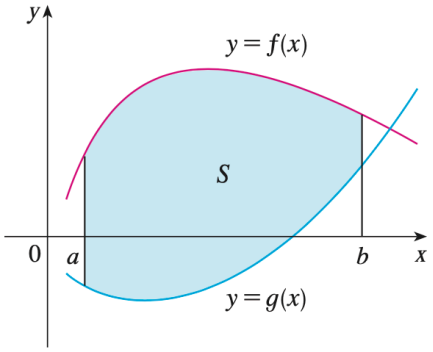
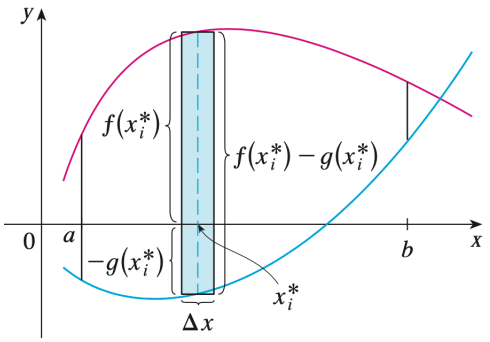
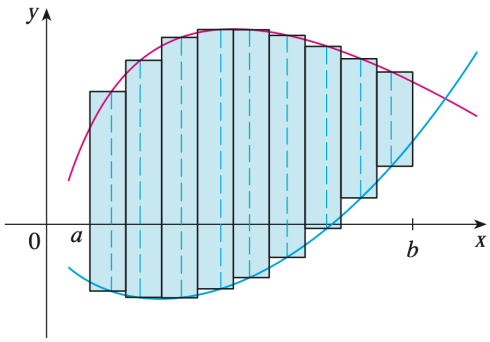


Figure 1: $S = \{(x,y) : a \leq x \leq b, g(x) \leq y \leq f(x)\}$.

will calculate this area as the limit of a Riemann sum. Partition the interval $[a,b]$ into n rectangles of width $\Delta x = \frac{b-a}{n}$. Choose a tagging $\tau = \{x_i^*\}$ of the partition, so that each rectangle has height $f(x_i^*) - g(x_i^*)$, as shown below:



(a) Typical rectangle



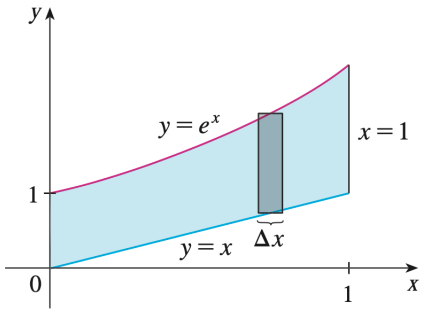
(b) Approximating rectangles

Since f and g are both continuous, they are integrable on $[a,b]$, so the area A of the region S is given by

$$\boxed{A = \lim_{n\to\infty} \sum_{j=1}^{n} (f(x_i^*) - g(x_i^*))\,\Delta x = \int_a^b (f(x) - g(x))dx}.$$

Notice that in the special case where $g(x) = 0$, S is the region under the graph of f and the formula above reduces to our old definition: $A = \int_a^b f(x)dx$.

1. Find the area of the region bounded by $y = e^x$, bounded below by $y = x$, and bounded on the sides by $x = 0$ and $x = 1$.

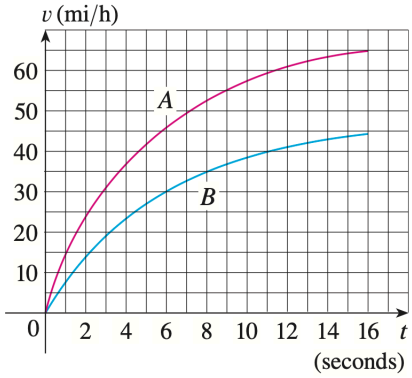


2. Sketch the region enclosed by the parabolas $y = x^2$ and $y = 2x - x^2$ and find the area of this region.

3. The figure below shows velocity curves for two cars, A and B, that start side by side and move along the same road.

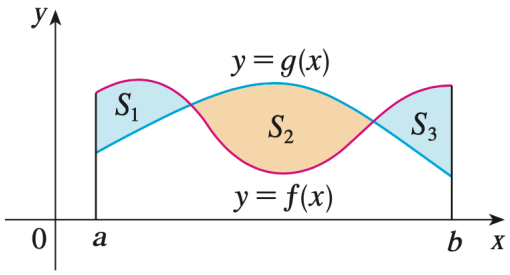
   (a) What does the area between the curves represent?

   (b) Use the Midpoint Rule to estimate it.



2 Curves that Cross

If we wish to find the area between the curves $y = f(x)$ and $y = g(x)$ where $f(x) \geq g(x)$ for some values of x but $g(x) \geq f(x)$ for other values of x, then we split the region S into several regions $S_1, S_2, \ldots$ with areas $A_1, A_2, \ldots$, as shown below.



We then define the area A of the region S to be the sum of the areas of the smaller regions $S_1, S_2, \ldots$, so that

$$A = A_1 + A_2 + \cdots + A_n.$$

Since

$$|f(x) - g(X)| = \begin{cases} f(x) - g(x) & \text{when } f(x) \geq g(x), \\ g(x) - f(x) & \text{when } g(x) \geq f(x), \end{cases}$$

we have the following expression for A:

$$\boxed{A = \sum_{i=1}^{n} \int_{x_i}^{x_{i+1}} |f(x) - g(x)| dx},$$

where x_i are the points where the graphs cross.

4. Sketch the region bounded by the curves $y = \sin(x)$ and $y = \cos(x)$ between $x = 0$ and $x = \frac{\pi}{2}$, and find the area of this region.

3 Regions between two functions of y

Some regions are best treated by regarding x as a function of y. If a region is bounded by curves with equations $x = f(y), x = g(y), y = c$, and $y = d$, where f and g are continuous and $f(y) \geq g(y)$ for $c \leq y \leq d$ (as shown below), then its area is

$$\boxed{A = \int_c^d (f(y) - g(y))dy}.$$

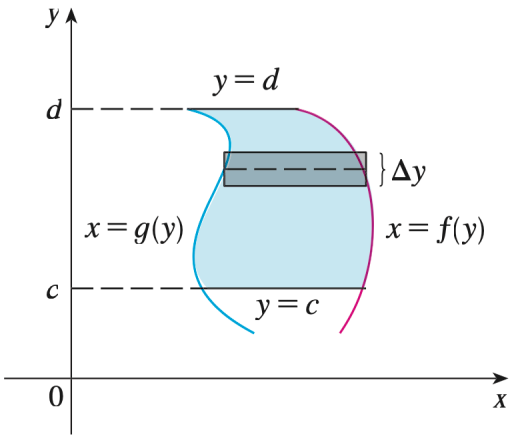


Figure 2: $S = \{(x,y) : g(y) \leq x \leq f(y), c \leq y \leq d\}$.

5. Sketch the region enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$, and calculate the area of this region.